PINK REPORT

Routine primary-tumor resection **should not be offered as standard care** for de novo metastatic HR+/HER2-negative breast cancer because **prospective randomized evidence does not show an overall survival benefit**; consider locoregional surgery **only in highly selected oligometastatic patients** (e.g., limited sites, excellent response to initial systemic therapy) after multidisciplinary discussion, primarily for **local control** (Tosello; Shien; NCCN).

This table summarizes evidence on whether resection of the primary breast tumor improves outcomes in de novo metastatic breast cancer. Across guideline and randomized-trial evidence, surgery **improves local control** but **does not consistently improve overall survival**, with any apparent subgroup benefits remaining exploratory (Tosello; Shien; Zhou; Villacampa; NCCN).

References supporting summary ordered by strength of evidence:

Rank	Level of Evidence	Study Title	Authors	Journal (Year)	Notes
🟩 1	**Guideline**	*Breast Cancer (NCCN Guidelines)*	National Comprehensive Cancer Network (NCCN)	NCCN (2026)	States routine locoregional treatment of the primary tumor **has not been shown to improve survival** in prospective trials; allows consideration of aggressive multimodality therapy in **selected de novo oligometastatic** cases with favorable features and response to systemic therapy.
🟢 2	**Clinical Trial Meta-analysis**	*Breast Surgery for Metastatic Breast Cancer*	Tosello G, Riera R, Torloni MR, et al.	Cochrane Database Syst Rev (2025)	Systematic review of **5 RCTs (n=1,368)**: **no OS benefit** with surgery (HR 0.89, 95% CI 0.75–1.05); improved **local control**; luminal/HR+ subgroup signal exploratory.
🟢 3	**Clinical Trial Meta-analysis**	*The Role of Locoregional Surgery in De Novo Stage IV Breast Cancer: A Meta-Analysis of Randomized Controlled Trials*	Zhou W, Yue Y, Xiong J, Li W, Zeng X	Cancer Treatment Reviews (2024)	Meta-analysis of **6 RCTs**: reports modest OS association (HR 0.86, 95% CI 0.77–0.96); larger effects reported in HR+ and HER2− subgroups—interpretation limited by between-trial heterogeneity and subgroup analyses.

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🟢 4	**Clinical Trial Meta-analysis**	*Impact of Primary Breast Surgery on Overall Survival of Patients With De Novo Metastatic Breast Cancer: A Systematic Review and Meta-Analysis*	Villacampa G, Papakonstantinou A, Fredriksson I, Matikas A	The Oncologist (2024)	Meta-analysis finds **no OS benefit overall** (HR 0.93, 95% CI 0.76–1.14); suggests possible benefit in **younger/premenopausal** subgroup—exploratory.
🟨🟢 5	**Phase 3 (P3) Trial**	*Primary Tumour Resection Plus Systemic Therapy Versus Systemic Therapy Alone in Metastatic Breast Cancer (JCOG1017, PRIM-BC): A Randomised Clinical Trial*	Shien T, Hara F, Aogi K, et al.	British Journal of Cancer (2025)	Randomized trial (**n=407**) after initial systemic therapy without progression: **no significant OS difference**; **better local relapse-free survival** with surgery; subgroup signals (e.g., premenopausal/single-organ) not definitive.
🟨 6	**Phase 2 (P2) Trial**				**No eligible Phase 2 trial evidence provided in the source material.**
🟧 7	**Phase 1 (P1) Trial**				**No eligible Phase 1 trial evidence provided in the source material.**
🟧🔴 8	**Large Cohort Study**	*Locoregional Therapy of the Primary Tumour in De Novo Stage IV Breast Cancer in 216,066 Patients: A Meta-Analysis*	Gera R, Chehade HELH, Wazir U, et al.	Scientific Reports (2020)	Pooled observational data (very large N) can be affected by **selection bias/confounding** (e.g., healthier/limited-burden patients more likely to receive surgery); supportive for hypothesis generation, not definitive for causality.
🟥 9	**Cohort Study**	*Primary-Site Local Therapy for Patients*	Khan SA, Schuetz S, Hosseini O	Annals of Surgical	Narrative/educational review (not primary cohort data): synthesizes trial and

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		With De Novo Metastatic Breast Cancer: An Educational Review		Oncology (2022)	observational findings; generally emphasizes **no routine OS benefit** and the role of **careful patient selection** and multidisciplinary decision-making.
🟫 10	**Animal Model**				**No animal-model evidence provided in the source material.**